

A Data for BU instances

Each job set requires a list of the jobs, as detailed in Table 12. For each of these jobs the machine processing each operation is specified and the number in brackets is the processing time. The order in which the machines, and respective processing times, are given corresponds to the order in which they must be processed. The travelling time between the LU area and the machines, and among the machines are reported in Table 13.

Table 12 Job Sets: sequence, machines, and processing times (Bilge and Ulusoy, 1995).

Job Set 1					Job Set 2				
Job 1	M1(8);	M2(16);	M4(12)		Job 1	M1(10);	M4(18)		
Job 2	M1(20);	M3(10);	M2(18)		Job 2	M2(10);	M4(18)		
Job 3	M3(12);	M4(8);	M1(15)		Job 3	M1(10);	M3(20)		
Job 4	M4(14);	M2(18)			Job 4	M2(10);	M3(15);	M4(12)	
Job 5	M3(10);	M1(15)			Job 5	M1(10);	M2(15);	M4(12)	
					Job 6	M1(10);	M2(15);	M3(12)	
Job Set 3					Job Set 4				
Job 1	M1(16);	M3(15)			Job 1	M4(11);	M1(10);	M2(7)	
Job 2	M2(18);	M4(15)			Job 2	M3(12);	M2(10);	M4(8)	
Job 3	M1(20);	M2(10)			Job 3	M2(7);	M3(10);	M1(9);	M3(8)
Job 4	M3(15);	M4(10)			Job 4	M2(7);	M4(8);	M1(12);	M2(6)
Job 5	M1(8);	M2(10);	M3(15);	M4(17)	Job 5	M1(9);	M2(7);	M4(8);	M2(10);
Job 6	M2(10);	M3(15);	M4(8);	M1(15)					M3(8)
Job Set 5					Job Set 6				
Job 1	M1(6);	M2(12);	M4(9)		Job 1	M1(9);	M2(11);	M4(7)	
Job 2	M1(18);	M3(6);	M2(15)		Job 2	M1(19);	M2(20);	M4(13)	
Job 3	M3(9);	M4(3);	M1(12)		Job 3	M2(14);	M3(20);	M4(9)	
Job 4	M4(6);	M2(15)			Job 4	M2(14);	M3(20);	M4(9)	
Job 5	M3(3);	M1(9)			Job 5	M1(11);	M3(16);	M4(8)	
					Job 6	M1(10);	M3(12);	M4(10)	
Job Set 7					Job Set 8				
Job 1	M1(6);	M4(6)			Job 1	M2(12);	M3(21);	M4(11)	
Job 2	M2(11);	M4(9)			Job 2	M2(12);	M3(21);	M4(11)	
Job 3	M2(9);	M4(7)			Job 3	M2(12);	M3(21);	M4(11)	
Job 4	M3(16);	M4(7)			Job 4	M2(12);	M3(21);	M4(11)	
Job 5	M1(9);	M3(18)			Job 5	M1(10);	M2(14);	M3(18);	M4(9)
Job 6	M2(13);	M3(19);	M4(6)		Job 6	M1(10);	M2(14);	M3(18);	M4(9)
Job 7	M1(10);	M2(9);	M3(13)						
Job 8	M1(11);	M2(9);	M4(8)						
Job Set 9					Job Set 10				
Job 1	M3(9);	M1(12);	M2(9);	M4(6)	Job 1	M1(11);	M3(19);	M2(16);	M4(13)
Job 2	M3(16);	M2(11);	M4(9)		Job 2	M2(21);	M3(16);	M4(14)	
Job 3	M1(21);	M2(18);	M4(7)		Job 3	M3(8);	M2(10);	M1(14);	M4(9)
Job 4	M2(20);	M3(22);	M4(11)		Job 4	M2(13);	M3(20);	M4(10)	
Job 5	M3(14);	M1(16);	M3(13);	M4(9)	Job 5	M1(9);	M3(16);	M4(18)	
					Job 6	M2(19);	M1(21);	M3(11);	M4(15)

Table 13 Travelling times in the layouts with four machines (Bilge and Ulusoy, 1995).

Layout 1	LU	M1	M2	M3	M4	Layout 2	LU	M1	M2	M3	M4
LU	0	6	8	10	12	LU	0	4	6	8	6
M1	12	0	6	8	10	M1	6	0	2	4	2
M2	10	6	0	6	8	M2	8	12	0	2	4
M3	8	8	6	0	6	M3	6	10	12	0	2
M4	6	10	8	6	0	M4	4	8	10	12	0
Layout 3	LU	M1	M2	M3	M4	Layout 4	LU	M1	M2	M3	M4
LU	0	2	4	10	12	LU	0	4	8	10	14
M1	12	0	2	8	10	M1	18	0	4	6	10
M2	10	12	0	6	8	M2	20	14	0	8	6
M3	4	6	8	0	2	M3	12	8	6	0	6
M4	2	4	6	12	0	M4	14	14	12	6	0

B Data for HK instances

The 23 instances in Set 2 are divided into four subsets; six small-sized instances, designated as P1_XX_YY, nine medium-sized instances, designated as P2_XX_YY, five medium-sized instances, designated as P2*0.5_XX_YY, and the forth subset: P1_tkl1, P2_tkl1, and P2_tkl1.

Each problem instance requires a list of jobs, as listed in Table 14, where each job has 6 (in P1 instances) or 10 (in P2 instances) operations, each of which with a specific machine and a processing time in brackets. For the five medium-sized instances start with P2*0.5, the processing times are the closest integer greater than or equal to the halved processing times of set P2 in Table 14.

Moreover, and as specified in the instances' names XX and YY indicate the corresponding loaded and empty travel times by the vehicle, respectively. Note that while the loaded travel times are job and operation dependent the empty travel times are machine dependent. The loaded travel times are defined as D1, as reported in Table 15; and then D2, D3 and D5 are defined as D1 times 2, 3, or 5, respectively. The empty travel times are defined as d1 in Table 15. Note that for some of the problem instances for which the loaded travel times is indicated as T1, T2, T3, and T5, all loaded travel times (regardless of the jobs and operations) are a constant value of 1, 2, 3, or 5, respectively. Similarly, all the empty travel times (between any pair of machines) in t0, t1, and t2, are defined as a constant value of 0, 1, and 2, respectively. Moreover, the loaded and empty travel times for the three instances, i.e., P1_tkl1, P2_tkl1, and P2_tkl2, are defined as reported in Table 16.

As an example, consider instance P2_D5_t2, for which the job set data is presented in Table 14: Job set P2, the loaded travel time can be calculated from Table 15: D1 times 5, and empty travel times are defined as a constant value of 2.

Table 14 Job Sets: sequence, machines, and processing times (Hurink and Knust, 2005).

Jobset P1										
Job 1	M3(1);	M1(3);	M2(6);	M4(7);	M6(3);	M5(6)				
Job 2	M2(8);	M3(5);	M5(10);	M6(10);	M1(10);	M4(4)				
Job 3	M3(5);	M4(4);	M6(8);	M1(9);	M2(1);	M5(7)				
Job 4	M2(5);	M1(5);	M3(5);	M4(3);	M5(8);	M6(9)				
Job 5	M3(9);	M2(3);	M5(5);	M6(4);	M1(3);	M4(1)				
Job 6	M2(3);	M4(3);	M6(9);	M1(10);	M5(4);	M3(1)				
Jobset P2										
Job 1	M1(29);	M2(78);	M3(9);	M4(36);	M5(49);	M6(11);	M7(62);	M8(56);	M9(44);	M10(21)
Job 2	M1(43);	M3(90);	M5(75);	M10(11);	M4(69);	M2(28);	M7(46);	M6(46);	M8(72);	M9(30)
Job 3	M2(91);	M1(85);	M4(39);	M3(74);	M9(90);	M6(10);	M8(12);	M7(89);	M10(45);	M5(33)
Job 4	M2(81);	M3(95);	M1(71);	M5(99);	M7(9);	M9(52);	M8(85);	M4(98);	M10(22);	M6(43)
Job 5	M3(14);	M1(6);	M2(22);	M6(61);	M4(26);	M5(69);	M9(21);	M8(49);	M10(72);	M7(53)
Job 6	M3(84);	M2(2);	M6(52);	M4(95);	M9(48);	M10(72);	M1(47);	M7(65);	M5(6);	M8(25)
Job 7	M2(46);	M1(37);	M4(61);	M3(13);	M7(32);	M6(21);	M10(32);	M9(89);	M8(30);	M5(55)
Job 8	M3(31);	M1(86);	M2(46);	M6(74);	M5(32);	M7(88);	M9(19);	M10(48);	M8(36);	M4(79)
Job 9	M1(76);	M2(69);	M4(76);	M6(51);	M3(85);	M10(11);	M7(40);	M8(89);	M5(26);	M9(74)
Job 10	M2(85);	M1(13);	M3(61);	M7(7);	M9(64);	M10(76);	M6(47);	M4(52);	M5(90);	M8(45)

Table 15 Travelling times (loaded and empty) for instances P1_XX_YY and P2_XX_YY in layouts with 6 and 10 machines (Hurink and Knust, 2005).

P1_XX_YY:

Loaded travelling times (D1)							Empty travelling times (d1)						
Job _i	O _{i1}	O _{i2}	O _{i3}	O _{i4}	O _{i5}	O _{i6}	M1	M2	M3	M4	M5	M6	
Job 1	0	2	1	2	2	1	M1	0	1	2	3	4	5
Job 2	0	1	2	1	5	3	M2	1	0	1	2	3	4
Job 3	0	1	2	5	1	3	M3	2	1	0	1	2	3
Job 4	0	1	2	1	1	1	M4	3	2	1	0	1	2
Job 5	0	1	3	1	5	3	M5	4	3	2	1	0	1
Job 6	0	2	2	5	4	2	M6	5	4	3	2	1	0

P2_XX_YY:

Loaded travelling times (D1)										Empty travelling times (d1)											
Job _i	O _{i1}	O _{i2}	O _{i3}	O _{i4}	O _{i5}	O _{i6}	O _{i7}	O _{i8}	O _{i9}	O _{i10}	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	
Job 1	0	1	1	1	1	1	1	1	1	1	M1	0	1	2	3	4	5	6	7	8	9
Job 2	0	2	2	5	6	2	5	1	2	1	M2	1	0	1	2	3	4	5	6	7	8
Job 3	0	1	3	1	6	3	2	1	3	5	M3	2	1	0	1	2	3	4	5	6	7
Job 4	0	1	2	4	2	2	1	4	6	4	M4	3	2	1	0	1	2	3	4	5	6
Job 5	0	2	1	4	2	1	4	1	2	3	M5	4	3	2	1	0	1	2	3	4	5
Job 6	0	1	4	2	5	1	9	6	2	3	M6	5	4	3	2	1	0	1	2	3	4
Job 7	0	1	3	1	4	1	4	1	1	3	M7	6	5	4	3	2	1	0	1	2	3
Job 8	0	2	1	4	1	2	2	1	2	4	M8	7	6	5	4	3	2	1	0	1	2
Job 9	0	1	2	2	3	7	3	1	3	4	M9	8	7	6	5	4	3	2	1	0	1
Job 10	0	1	2	4	2	1	4	2	1	3	M10	9	8	7	6	5	4	3	2	1	0

Table 16 Travelling times (loaded and empty) for instance P1_tkl.1, P2_tkl.1, and P2_tkl.2 in layouts with 6, and 10 machines (Hurink and Knust, 2005).

P1.tkl.1:																					
Loaded travelling times																					
Job_i	O_{i1}	O_{i2}	O_{i3}	O_{i4}	O_{i5}	O_{i6}															
Job 1	0	6	3	3	6	3															
Job 2	0	2	1	5	6	6															
Job 3	0	3	6	6	3	1															
Job 4	0	5	4	3	2	5															
Job 5	0	4	1	5	6	6															
Job 6	0	3	6	6	4	1															
P2.tkl.1:																					
Loaded travelling times											Empty travelling times										
Job_i	O_{i1}	O_{i2}	O_{i3}	O_{i4}	O_{i5}	O_{i6}	O_{i7}	O_{i8}	O_{i9}	O_{i10}		M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
Job 1	0	5	3	2	1	5	6	3	4	4	M1	0	3	3	1	2	5	2	2	1	2
Job 2	0	7	3	2	4	6	5	6	4	4	M2	2	0	2	1	1	3	2	1	2	2
Job 3	0	3	4	5	4	5	4	4	3	3	M3	2	2	0	2	1	3	1	1	2	2
Job 4	0	3	2	3	4	3	4	6	3	6	M4	3	2	2	0	1	4	2	1	2	1
Job 5	0	2	5	4	3	1	3	4	2	2	M5	3	2	2	1	0	4	3	1	2	1
Job 6	0	7	4	3	2	4	4	2	1	2	M6	1	1	2	2	1	0	3	2	2	2
Job 7	0	3	4	5	4	6	4	5	4	5	M7	3	3	3	1	1	4	0	1	2	2
Job 8	0	2	5	4	2	4	3	4	5	6	M8	2	4	3	3	2	3	3	0	1	2
Job 9	0	5	5	4	2	2	2	3	5	3	M9	1	3	2	2	1	4	3	2	0	2
Job 10	0	3	7	4	3	4	6	3	1	2	M10	3	2	1	2	1	4	2	2	3	0
P2.tkl.2:																					
Loaded travelling times											Empty travelling times										
Job_i	O_{i1}	O_{i2}	O_{i3}	O_{i4}	O_{i5}	O_{i6}	O_{i7}	O_{i8}	O_{i9}	O_{i10}		M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
Job 1	0	7	3	6	3	4	5	3	1	6	M1	0	4	4	3	2	3	2	3	3	5
Job 2	0	6	6	9	1	3	5	1	5	1	M2	1	0	1	2	1	2	1	2	1	1
Job 3	0	2	6	1	4	1	5	3	6	3	M3	2	1	0	3	1	3	2	2	2	2
Job 4	0	3	4	7	8	4	6	7	5	1	M4	1	2	1	0	2	2	3	1	1	3
Job 5	0	4	7	4	6	3	3	6	7	4	M5	2	2	3	2	0	3	3	3	2	3
Job 6	0	2	4	6	2	6	2	8	5	7	M6	2	1	2	3	2	0	1	2	2	2
Job 7	0	2	6	1	2	1	5	2	6	2	M7	3	2	2	3	1	1	0	1	1	3
Job 8	0	4	7	4	6	8	4	6	1	7	M8	2	1	1	3	2	1	2	0	1	2
Job 9	0	7	5	3	4	6	4	3	2	3	M9	2	2	2	3	2	1	2	1	0	2
Job 10	0	2	6	2	4	6	1	6	3	7	M10	1	1	1	1	2	1	1	1	2	0

C Travelling times for SWV instances

For the 10 instances in Set 3 (designated as SWV01:10) the list of the jobs, each of which with a set of operations with a specific machine and processing time are available at the OR library³. Moreover, the travelling time are reported in Table 17 for layouts with 10 and 15 machines.

³ <http://people.brunel.ac.uk/~mastjjb/jeb/orlib/files/jobshop1.txt>

Table 17 Traveling times for layouts with 10 and 15 machines (Ham, 2020).

Traveling times for 10 machines layout																
	LU	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10					
LU	0	22	18	6	8	20	7	25	31	12	27					
M1	22	0	11	27	30	32	20	10	28	21	7					
M2	18	11	0	22	26	22	21	7	18	12	19					
M3	6	27	22	0	6	17	13	29	32	12	33					
M4	8	30	26	6	0	23	12	34	38	18	34					
M5	20	32	22	17	23	0	28	29	20	11	40					
M6	7	20	21	13	12	28	0	26	37	18	23					
M7	25	10	7	29	34	29	26	0	19	19	17					
M8	31	28	18	32	38	20	37	19	0	19	35					
M9	12	21	12	12	18	11	18	19	19	0	28					
M10	27	7	19	33	34	40	23	17	35	28	0					
Traveling times for 15 machines layout																
	LU	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15
LU	0	14	11	33	38	31	31	4	18	22	19	29	4	30	2	19
M1	14	0	11	27	30	32	20	10	28	21	7	22	17	34	16	20
M2	11	11	0	22	26	22	21	7	18	12	19	18	15	23	12	10
M3	33	27	22	0	6	17	13	29	32	12	33	6	37	23	34	16
M4	38	30	26	6	0	23	12	34	38	18	34	8	42	29	39	21
M5	31	32	22	17	23	0	28	29	20	11	40	20	34	6	31	12
M6	31	20	21	13	12	28	0	26	37	18	23	7	35	33	33	21
M7	4	10	7	29	34	29	26	0	19	19	17	25	8	28	6	17
M8	18	28	18	32	38	20	37	19	0	19	35	31	19	16	17	16
M9	22	21	12	12	18	11	18	19	19	0	28	12	26	15	23	3
M10	19	7	19	33	34	40	23	17	35	28	0	27	22	42	21	28
M11	29	22	18	6	8	20	7	25	31	12	27	0	33	25	31	15
M12	4	17	15	37	42	34	35	8	19	26	22	33	0	33	3	23
M13	30	34	23	23	29	6	33	28	16	15	42	25	33	0	30	14
M14	2	16	12	34	39	31	33	6	17	23	21	31	3	30	0	20
M15	19	20	10	16	21	12	21	17	16	3	28	15	23	14	20	0